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Power factor high-voltage end synchronous rectifier

Abstract

A power factor high-voltage end synchronous rectifier includes a bridge rectifying module, a power factor module and a high-voltage synchronous rectifier, whereby the high-voltage synchronous rectifier obtains the state of a second power outputted by the bridge rectifying module through a current detector and generates first and second control signals. The first control signal adjusts the on/off state of a first wide bandgap compound semiconductor of the power factor module, and the second control signal adjusts the on/off state of a second wide bandgap compound semiconductor of the power factor module, allowing the high-voltage synchronous rectifier to generate a plurality of control signals at the same time, so that the second power of the power module can be switched between continuous conduction mode (CCM), discontinuous conduction mode (DCM) and quasi resonant mode (QR) in a very short time (can be regarded as zero second in an ideal state).

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a power factor high-voltage end synchronous rectifier, and more particularly, to a power factor high-voltage end synchronous rectifier that is capable of reducing power consumption while being applicable in discontinuous conduction mode (DCM), continuous conduction mode (CCM) and quasi resonant (QR) mode.

2. Description of the Prior Art

(2) The technology of switching-mode power supply (SMPS) has been widely used in the production of power supplies for various electrical devices and electronic products. With the trend of electronic products becoming lighter, thinner and smaller, it is necessary to increase the power density of SMPS by increasing the switching frequency, in order to effectively reduce the size of the SMPS mechanism. In view of this, LLC resonant converters with Zero Voltage Switching (ZVS) and Zero Current Switching (ZCS) characteristics have been proposed.

(3) Power Factor Correction (PFC) is a circuit that can improve the power factor of the power supply end and is commonly used in the aforementioned SMPS. It can be mainly divided into passive PFC, active PFC, and dynamic PFC. Regardless of which type of PFC circuit, it outputs a single control signal. For today's high-voltage, high-frequency power inputs, and the need for multiple voltage outputs, even slight delays can lead to less than ideal power supply outputs. Especially at the output end, all PFC circuits use simple diodes as isolation, and losses increase as

the load increases.

(4) In view of the shortcomings in the prior art, the present disclosure is proposed to provide improvements that address these shortcomings.

SUMMARY OF THE INVENTION

(5) One main objective of the present disclosure is to provide a power factor high-voltage end synchronous rectifier capable of reducing power consumption.

(6) Another objective of the present disclosure is to provide a power factor high-voltage end synchronous rectifier that can be used in discontinuous conduction mode (DCM), continuous conduction mode (CCM) and quasi resonant (QR) mode.

(7) Yet another objective of the present disclosure is to provide a power factor high-voltage end synchronous rectifier that can be applied to different types of power supplies.

(8) In order to achieve the above objectives and efficacies, the technical means employed by the present disclosure may include: a bridge rectifying module, a power factor module and a high-voltage synchronous rectifier.

(9) The bridge rectifying module may include a positive output terminal and a negative output terminal for converting an alternating current (AC) first electrical energy into a direct current (DC) second electric energy, and the second electrical energy being output through the positive output terminal.

(10) The power factor module may be electrically connected to the bridge rectifying module and may include: a power factor correction (PFC) inductor, a first wide bandgap compound semiconductor, a second wide bandgap compound semiconductor and a current detector.

(11) The PFC inductor may be electrically connected to the positive output terminal of the bridge rectifying module.

(12) The first wide bandgap compound semiconductor may include a first gate, a first drain and a first source, the first drain being electrically connected with the PFC inductor, and the first source being electrically connected to the negative output terminal of the bridge rectifying module.

(13) The second wide bandgap compound semiconductor may include a second gate, a second drain and a second source, the second source being electrically connected to the first drain.

(14) The current detector may include a first pin, a second pin, a third pin and a fourth pin.

(15) The first pin may be electrically connected to the second drain and the third pin may be electrically connected to an overall output terminal.

(16) The high-voltage synchronous rectifier may include a first control terminal, a second control terminal, a first input terminal, and a second input terminal, wherein:

(17) The first control terminal may be electrically connected to the first gate of the first wide bandgap compound semiconductor, the second control terminal may be electrically connected to the second gate of the second wide bandgap compound semiconductor, the first input terminal may be electrically connected to the second pin of the current detector, and the second input terminal may receive a PWM signal.

(18) As a result, the first input terminal of the high-voltage synchronous rectifier obtains the state of the second electrical energy through the second pin of the current detector, and at the same time, the high-voltage synchronous rectifier obtains the PWM signal from the second input terminal, then the high-voltage synchronous rectifier generates a first control signal and a second control signal, the first control terminal transmits the first control signal to control the on or off state of the first wide bandgap compound semiconductor, and the second control terminal transmits the second control signal to control the on or off state of the second wide bandgap compound semiconductor, so that the second electrical energy of the bridge rectifying module is switched between continuous conduction mode (CCM) and discontinuous conduction mode (DCM), and quasi resonant mode (QR).

(19) Based on the structure above, the bridge rectifying module may include: a filter for receiving a first electrical energy from a predetermined external power source and eliminating ripples in the

first electrical energy; a full-wave rectifier electrically connected to the filter having the positive output terminal and the negative output terminal, the full-wave rectifier converting the AC first electrical energy into the DC second electrical energy, the second electrical energy being output through the positive output terminal

(20) Based on the structure above, the high-voltage synchronous rectifier may receive the PWM signal and the second electrical energy, and generate the first control signal to control the first gate and generate the second control signal to control the second gate in the following modes:

(21) When the PWM signal is at High (Hi) and the current detector is at Low (Lo), the first control signal at Hi is generated to control the first gate, turning the first wide bandgap semiconductor on, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off.

(22) When the PWM signal is at Hi and the current detector is at Hi, the first control signal at Hi is generated to control the first gate, turning the first wide bandgap semiconductor on, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off.

(23) Based on the structure above, the high-voltage synchronous rectifier may receive the PWM signal and the second electrical energy, and generate the first control signal to control the first gate and generate the second control signal to control the second gate in the following modes:

(24) When the PWM signal is at Lo and the current detector is at Lo, the first control signal at Lo is generated to control the first gate, turning the first wide bandgap semiconductor off, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off.

(25) When the PWM signal is at Lo and the current detector is at Hi, the first control signal at Lo is generated to control the first gate, turning the first wide bandgap semiconductor off, meanwhile, the second control signal at Hi is generated to control the second gate, turning the second wide bandgap semiconductor on.

(26) Based on the structure above, the current detector may be a resistive type, a magnetic type with an iron core, a magnetic type without an iron core, or a current comparator.

(27) Based on the structure above, the first or second wide bandgap compound semiconductor may be a metal-oxide-semiconductor field-effect transistor (MOSFET) made of silicon carbide (SiC) or a MOSFET made of gallium nitride (GaN).

(28) Based on the structure above, the PFC inductor may be a low impedance coil and a choke coil.

(29) The objectives, efficacies and features of the present disclosure can be more fully understood by referring to the drawing as follows:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a preferred embodiment of the present disclosure.

(2) FIG. 2 is a first waveform of a preferred embodiment of the present disclosure.

(3) FIG. 3 is a second waveform of a preferred embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(4) Referring to FIG. 1, the structure of the invention includes a bridge rectifying module 1, a power factor module 2, and a high-voltage synchronous rectifier 3.

(5) The bridge rectifying module 1 includes a filter 13 and a full-wave rectifier 14.

(6) The filter 13 is connected to an external power source and is used to receive and eliminate ripples and noise from a first electrical energy.

(7) The full-wave rectifier 14 has a positive output terminal 11 and a negative output terminal 12.

(8) The full-wave rectifier 14 converts the AC-type first electrical energy to a DC-type second

electrical energy and outputs it through the positive output terminal **11**.

(9) The power factor module **2** is connected to the bridge rectifying module **1** and includes a power factor correction (PFC) inductor **21**, a first wide bandgap compound semiconductor **22**, a second wide bandgap compound semiconductor **23** and a current detector **24**.

(10) The PFC inductor **21** is connected to the positive output terminal **11** of the full-wave rectifier **14**. In a preferred embodiment, the PFC inductor can be a low impedance coil or a choke coil.

(11) The first wide bandgap compound semiconductor **22** has a first gate **221**, a first drain **222**, and a first source **223**.

(12) The first drain **222** is electrically connected to the PFC inductor **21**, and the first source **223** is electrically connected to the negative output terminal **12** of the full-wave rectifier **14**.

(13) The second wide bandgap compound semiconductor **23** has a second gate **231**, a second drain **232**, and a second source **233**.

(14) The second source **233** is electrically connected to the first drain **222** of the first wide bandgap compound semiconductor **22**.

(15) In another preferred embodiment, the first and second wide bandgap compound semiconductors can be metal-oxide-semiconductor field-effect transistors (MOSFETs) made of silicon carbide (SiC) or gallium nitride (GaN).

(16) The current detector **24** has a first pin **241**, a second pin **242**, a third pin **243**, and a fourth pin **244**.

(17) The first pin **241** is electrically connected to the second drain **232** of the second wide bandgap compound semiconductor **23**, and the third pin **243** is electrically connected to the overall output terminal **4**.

(18) In a preferred embodiment, the current detector **24** can be resistive, magnetic with an iron core, magnetic without an iron core, or a current comparator.

(19) The high-voltage synchronous rectifier **3** has a first control terminal **31**, a second control terminal **32**, a first signal terminal **33**, and a second signal terminal **34**.

(20) The first control terminal **31** is electrically connected to the first gate **221** of the first wide bandgap compound semiconductor **22**.

(21) The second control terminal **32** is electrically connected to the second gate **231** of the second wide bandgap compound semiconductor **23**.

(22) The first signal terminal **33** is electrically connected to the second pin **242** of the current detector **24**.

(23) The second input terminal **34** receives a PWM signal.

(24) Thus, the first signal terminal **33** of the high-voltage synchronous rectifier **3** obtains the status of the current of the second electrical energy through the second pin **24** of the current detector **24**. The high-voltage synchronous rectifier **3** obtains the PWM signal from the second signal terminal **34**, and then generates a first control signal and a second control signal. The first control terminal **31** transmits the first control signal for controlling the on/off state of the first wide bandgap semiconductor **22**, and the second control terminal **32** transmits the second control signal for controlling the on/off state of the second wide bandgap semiconductor **23**, so that the second electrical energy of the bridge rectifying module **1** can switch between the continuous conduction mode (CCM), discontinuous conduction mode (DCM), and quasi resonant mode (QR).

(25) Referring to both FIGS. **1** and **2** at the same time. FIG. **2** shows the waveform of the present disclosure during operation. In a preferred embodiment, the structure of the present disclosure shown in FIG. **1** is connected to a conventional power supply circuit, and the "Input" in FIG. **2** represents an existing input control in the conventional power supply circuit. In addition, "PWM" in FIGS. **1** and **2** represents an existing PWM input signal in the current power supply circuit. FIG. **2** represents the operating state of the present disclosure under continuous conduction mode (CCM) after combining with the current power supply circuit.

(26) The high-voltage synchronous rectifier **3** receives the PWM signal and the second electrical

energy, and generates the first control signal to control the first gate **221** and generates the second control signal to control the second gate **231** in the following modes:

(27) When the PWM signal is at High (Hi) and the current detector is at Low (Lo), a Hi first control signal is generated to control the first gate **221**, turning the first wide bandgap semiconductor **22** on. At the same time, a Lo second control signal is generated to control the second gate **231**, turning the second wide bandgap semiconductor **23** off.

(28) When the PWM signal is at Hi and the current detector is at Hi, a Hi first control signal is generated to control the first gate **221**, turning the first wide bandgap semiconductor **22** on. At the same time, a Lo second control signal is generated to control the second gate **231**, turning the second wide bandgap semiconductor **23** off.

(29) When the current power supply circuit (which may be a controller, a signal generator, or another electronic circuit combination capable of generating a PWM signal) sends a PWM signal, the PWM signal is received by the second signal terminal **34** of the high-voltage synchronous rectifier **3**. At the same time, the state of the second electrical energy is obtained by the first signal terminal **33** of the high-voltage synchronous rectifier **3** (obtained from the second pin **242** of the current detector **24**), the high-voltage synchronous rectifier **3** then generates the first control signal and the second control signal depending on a combination of the PWM signal and the state of the second electrical energy. The first control signal is transmitted to the first wide bandgap semiconductor **22** through the first control terminal **31** of the high-voltage synchronous rectifier **3**, and the second control signal is transmitted to the second wide bandgap semiconductor **23** through the second control terminal **32** of the high-voltage synchronous rectifier **3**. In a preferred embodiment, the high-voltage synchronous rectifier **3** controls the first and second control signals, such that the first and second control signals are interlocked to prevent both from conducting at the same time.

(30) Referring to both FIGS. **1** and **3** at the same time, FIG. **3** shows the waveform of the present disclosure during operation. Descriptions that are the same as those described in the preferred embodiment above (with respect to FIGS. **1** and **2**) will not repeated again. FIG. **3** represents the operating states of the present disclosure under discontinuous conduction mode (DCM) and quasi resonant (QR) mode after combining with the current power supply circuit.

(31) The high-voltage synchronous rectifier **3** receives the PWM signal and the second electrical energy, and generates the first control signal to control the first gate **221** and generates the second control signal to control the second gate **231** in the following modes:

(32) When the PWM signal is at Lo and the current detector is at Lo, a Lo first control signal is generated to control the first gate **221**, turning the first wide bandgap semiconductor **22** off. At the same time, a Lo second control signal is generated to control the second gate **231**, turning the second wide bandgap semiconductor **23** off.

(33) When the PWM signal is at Lo and the current detector is at Hi, a Lo first control signal is generated to control the first gate **221**, turning the first wide bandgap semiconductor **22** off. At the same time, a Hi second control signal is generated to control the second gate **231**, turning the second wide bandgap semiconductor **23** on. It should be noted that, in this embodiment, descriptions related to the signal transmission and circuit connections of the current power supply circuit can be found to the relevant paragraphs of the preferred embodiment above (descriptions with respect to FIGS. **1** and **2**) and will not repeated again.

(34) In summary of the descriptions with respect to FIGS. **1** to **3** above, a truth table of the present disclosure can be as follows:

(35) TABLE-US-00001

PWM	IN	Hi	Hi	Lo	Lo	Current Detector	Lo	Hi	Lo	Hi	First Control Signal	Hi	Hi	Lo	Lo	Second Control Signal	Lo	Lo	Lo	Hi	First Wide Bandgap Compound	ON	ON	OFF	OFF	Semiconductor	Second Wide Bandgap Compound	OFF	OFF	OFF	ON	Semiconductor
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(36) The present disclosure is further illustrated with an example: Under the traditional architecture, the loss of a boost diode is $((1-D)*V_f*I_f)$. By reducing the loss and increasing

efficiency through the architecture of the present disclosure: V_f : Forward voltage drop; I_f : Forward current; R_{dson} : Conductive impedance of the first wide bandgap compound semiconductor **22** or the second wide bandgap compound semiconductor **23** mentioned before; Then, the loss generated by V_f , I_f , and R_{dson} is $[(1-D)*R_{dson}*I_f]$; For example: For a power of 300 W, $I_f \approx 300/90$ Vac (maximum effective current) ≈ 3.33 A; $D=0.4$ duty cycle, $V_f \approx 1.2$ V; Therefore, $(1-0.4)*1.2*3.33$ A = 2.39 W;

(37) With the structure of the present disclosure, $(1-0.4)*0.055*3.332$ A = 0.3659 W, increasing efficiency by at least 0.6%. The above is just an example of how efficiency improvement is calculated, and the efficiency improvement achieved by the present disclosure is not limited to 0.6%.

(38) In view of this, the present disclosure is submitted to be novel and non-obvious and a patent application is hereby filed in accordance with the patent law. It should be noted that the descriptions given above are merely descriptions of preferred embodiments of the present disclosure, various changes, modifications, variations or equivalents can be made to the invention without departing from the scope or spirit of the invention. It is intended that all such changes, modifications and variations fall within the scope of the following appended claims and their equivalents.

Claims

1. A power factor high-voltage end synchronous rectifier, comprising: a bridge rectifying module including a positive output terminal and a negative output terminal for converting an alternating current (AC) first electrical energy into a direct current (DC) second electric energy, and the second electrical energy being output through the positive output terminal; a power factor module electrically connected to the bridge rectifying module including: a power factor correction (PFC) inductor electrically connected to the positive output terminal of the bridge rectifying module; a first wide bandgap compound semiconductor having a first gate, a first drain and a first source, the first drain being electrically connected with the PFC inductor, and the first source being electrically connected to the negative output terminal of the bridge rectifying module; a second wide bandgap compound semiconductor having a second gate, a second drain and a second source, the second source being electrically connected to the first drain; a current detector having a first pin, a second pin, a third pin and a fourth pin, the first pin being electrically connected to the second drain, the third pin being electrically connected to an overall output terminal; and, a high-voltage synchronous rectifier having a first control terminal, a second control terminal, a first input terminal, and a second input terminal, wherein: the first control terminal is electrically connected to the first gate of the first wide bandgap compound semiconductor, the second control terminal is electrically connected to the second gate of the second wide bandgap compound semiconductor, the first input terminal is electrically connected to the second pin of the current detector, and the second input terminal receives a PWM signal, whereby the first input terminal of the high-voltage synchronous rectifier obtains the state of the second electrical energy through the second pin of the current detector, and at the same time, the high-voltage synchronous rectifier obtains the PWM signal from the second input terminal, the high-voltage synchronous rectifier then generates a first control signal and a second control signal, the first control terminal transmits the first control signal to control the on or off state of the first wide bandgap compound semiconductor, and the second control terminal transmits the second control signal to control the on or off state of the second wide bandgap compound semiconductor, so that the second electrical energy of the bridge rectifying module is switched between continuous conduction mode (CCM) and discontinuous conduction mode (DCM), and quasi resonant mode (QR).

2. The power factor high-voltage end synchronous rectifier of claim 1, wherein the bridge rectifying module includes: a filter for receiving the first electrical energy from a predetermined

external power source and eliminating ripples in the first electrical energy; a full-wave rectifier electrically connected to the filter having the positive output terminal and the negative output terminal, the full-wave rectifier converting the AC first electrical energy into the DC second electrical energy, the second electrical energy being output through the positive output terminal.

3. The power factor high-voltage end synchronous rectifier of claim 1, wherein the high-voltage synchronous rectifier receives the PWM signal and the second electrical energy, and generates the first control signal to control the first gate and generates the second control signal to control the second gate in the following modes: when the PWM signal is at High (Hi) and the current detector is at Low (Lo), the first control signal at Hi is generated to control the first gate, turning the first wide bandgap semiconductor on, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off; when the PWM signal is at Hi and the current detector is at Hi, the first control signal at Hi is generated to control the first gate, turning the first wide bandgap semiconductor on, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off.

4. The power factor high-voltage end synchronous rectifier of claim 1, wherein the high-voltage synchronous rectifier receives the PWM signal and the second electrical energy, and generates the first control signal to control the first gate and generates the second control signal to control the second gate in the following modes: when the PWM signal is at Lo and the current detector is at Lo, the first control signal at Lo is generated to control the first gate, turning the first wide bandgap semiconductor off, meanwhile, the second control signal at Lo is generated to control the second gate, turning the second wide bandgap semiconductor off; when the PWM signal is at Lo and the current detector is at Hi, the first control signal at Lo is generated to control the first gate, turning the first wide bandgap semiconductor off, meanwhile, the second control signal at Hi is generated to control the second gate, turning the second wide bandgap semiconductor on.

5. The power factor high-voltage end synchronous rectifier of claim 1, wherein the current detector is at least one selected from the group consisting of a resistive type, a magnetic type with an iron core, a magnetic type without an iron core, and a current comparator.

6. The power factor high-voltage end synchronous rectifier of claim 1, wherein the first or second wide bandgap compound semiconductor is at least one selected from the group consisting of a metal-oxide-semiconductor field-effect transistor (MOSFET) made of silicon carbide (SiC) and a MOSFET made of gallium nitride (GaN).

7. The power factor high-voltage end synchronous rectifier of claim 1, wherein the PFC inductor is at least one selected from the group consisting of a low impedance coil and a choke coil.
